1. Research Paper – EDA, Insights and Recommendations

The first deliverable is a comprehensive **Research Paper** documenting the complete analytical study performed on the Nassau Candy Distributor dataset.

The research paper provides a structured explanation of the business problem, dataset, analytical methodology, findings, and recommendations.

Contents of the Research Paper

The research paper includes the following major sections:

1.1 Introduction

Provides an overview of the project and explains the importance of analysing product sales, gross profit, costs, margins, and product-level profitability.

1.2 Problem Statement

The analysis focuses on understanding:

- Which products generate higher profitability?
- Which products have stronger or weaker margins?
- How do different divisions perform?
- Which products may require margin or cost review?
- How concentrated are revenue and profit among products?

1.3 Objectives

The research paper documents the objectives of the analysis, including:

- Analysing product-level profitability
- Calculating gross margin and profit per unit
- Comparing sales and profit across divisions
- Identifying potential margin-risk products
- Analysing cost versus sales relationships
- Measuring revenue and profit concentration
- Providing data-driven management recommendations

1.4 Dataset Description

The dataset contains information relating to orders, customers, products, sales, units, costs, gross profit, dates, and divisions.

Important fields used in the analysis include:

- Order Date
- Product Name
- Sales
- Units
- Gross Profit
- Cost
- Division
- Gross Margin (%)
- Profit per Unit

1.5 Data Cleaning and Preparation

The research paper documents the preparation of the dataset, including:

- Converting order dates into appropriate date format
- Converting financial and quantity fields into numeric values
- Handling invalid or missing values
- Creating calculated analytical fields
- Validating sales, cost, units, and gross-profit values

1.6 Exploratory Data Analysis

EDA is performed to understand the overall structure and characteristics of the dataset.

The analysis examines:

- Total sales
- Total gross profit
- Total units
- Overall gross margin
- Product-level sales
- Product-level profit
- Product-level margins
- Division-level sales
- Division-level profit

1.7 Product Profitability Analysis

Product-level analysis evaluates:

- Gross Profit
- Gross Margin (%)
- Profit per Unit
- Revenue Contribution
- Profit Contribution

Products are compared to identify differences in profitability and contribution to the overall business.

1.8 Division Performance Analysis

The analysis compares the major divisions using:

- Total Sales
- Gross Profit
- Average Gross Margin

This provides a high-level view of the contribution and profitability characteristics of each division.

1.9 Cost and Margin Diagnostics

The project examines the relationship between:

- Cost
- Sales
- Gross Margin

The analysis also identifies products falling below the selected margin threshold for further review. The threshold is treated as a **dashboard screening parameter**, rather than as a universal industry benchmark.

1.10 Pareto and Concentration Analysis

Pareto analysis is performed separately for:

- Revenue
- Gross Profit

Cumulative contribution is calculated to determine how many products contribute approximately 80% of revenue and profit.
This provides an indication of the concentration of business performance among products.

1.11 Key Insights

The research paper consolidates the analytical findings into meaningful business observations.
For example:

- Certain products contribute substantially to overall revenue and profit.
- Product margins vary considerably.
- Division-level profitability differs.
- Some products require further margin or cost review.
- Revenue and profit are concentrated among a relatively small number of products.

1.12 Recommendations

Based on the analysis, recommendations include:

- Periodically monitor high-profit-contributing products.
- Review products with comparatively low margins.
- Examine pricing and discount structures for margin-sensitive products.
- Review procurement costs for cost-heavy products.
- Monitor products contributing significantly to total revenue and profit.
- Use the dashboard for periodic profitability monitoring.
- Conduct further investigation before making pricing, procurement, or product-level decisions.

2. Streamlit Dashboard – Live Analytics

The second major deliverable is an **interactive Streamlit dashboard** developed using Python, Pandas, and Plotly.

The dashboard converts the analytical calculations into an interactive visual interface that allows users to explore profitability performance without directly working with Python code.

2.1 Dashboard Purpose

The dashboard provides a single interface for monitoring:

- Product profitability
- Product margins
- Profit contribution
- Division performance
- Cost and margin relationships
- Margin risks
- Revenue concentration
- Profit concentration

2.2 Interactive Filters

The dashboard provides four primary filters.

Date Range
Allows the user to analyse the business performance for a selected period.

Division
Allows users to select one or more divisions for comparison.

Margin Threshold
Allows users to define a margin screening threshold.
Products below the selected threshold are identified as potential margin-risk products.

Product Search
Allows users to search for a specific product and examine its profitability information.

2.3 Executive Summary KPIs

The dashboard displays four primary KPIs:

Total Sales
Shows the total sales generated by the selected data.

Gross Profit
Shows the total gross profit generated by the selected products or divisions.

Total Units
Shows the total number of units represented in the selected data.

Gross Margin
Calculated as:
Gross Margin(%)=Gross ProfitSales×100Gross\ Margin(\%) = \frac{Gross\ Profit}{Sales}\times100

These KPIs provide a quick overview of the selected business performance.

2.4 Module 1 – Product Profitability Overview

This module provides detailed product-level profitability analysis.

It includes:

Product-Level Margin Leaderboard
Displays products according to their gross margin percentage.
This allows users to identify products with relatively higher or lower margins.

Profit Contribution by Product
Shows how much each product contributes to total gross profit.
This helps identify products that make a significant contribution to overall profitability.

2.5 Module 2 – Division Performance Dashboard

This module evaluates performance across divisions.

Revenue vs Profit by Division
Compares sales and gross profit for each division.
This allows users to understand the relative scale and profitability contribution of different divisions.

Margin Distribution by Division
A boxplot is used to show the distribution of gross margins within each division.
The selected margin threshold is displayed as a reference line.

2.6 Module 3 – Cost vs Margin Diagnostics

This module focuses on identifying potential areas requiring further investigation.

Cost vs Sales
A scatter plot is used to examine the relationship between product cost and sales.
The visualization helps identify products or patterns that may warrant further analysis.
The chart is used to identify **patterns for investigation** and does not by itself establish that cost causes lower or higher sales.

Margin Risk Flags
A small analytical table identifies products based on the selected margin threshold.
Products are classified as:

- **Below Threshold**
- **Healthy**

The table contains:

- Product Name
- Sales
- Gross Profit
- Cost
- Margin
- Risk Flag

2.7 Module 4 – Profit Concentration Analysis

This module evaluates the concentration of business performance among products.

Revenue Pareto Analysis
Products are ranked by sales from highest to lowest.
The cumulative sales percentage is then calculated to identify the products contributing approximately 80% of revenue.

Revenue Dependency

The dashboard displays:

- Number of products required to generate approximately 80% of revenue
- Percentage of total products represented by those products

Profit Pareto Analysis

Products are ranked according to gross profit.

The cumulative profit percentage is calculated to determine the products contributing approximately 80% of profit.

Profit Dependency

The dashboard displays:

- Number of products required to generate approximately 80% of profit
- Percentage of total products represented by those products

2.8 Dashboard Technology

The dashboard is developed using:

Technology	Purpose
Python	Data processing and application development
Pandas	Data cleaning and analysis
Streamlit	Interactive dashboard
Plotly	Interactive charts and visualizations
CSV	Processed analytical dataset

2.9 Dashboard Output

The final dashboard will be submitted as:

```
02_Streamlit_Dashboard/
│
├── Ncdp.py
│
├── Nassau_Candy_Processed.csv
│
└── assets/
    └── candy_header.png
```

The dashboard can be executed using:

```
streamlit run Ncdp.py
```

3. Executive Summary for Government Stakeholders

The third deliverable is a concise **Executive Summary** prepared for senior management and government stakeholders.

Unlike the research paper, which contains detailed technical methodology, the executive summary focuses on **business findings, performance indicators, areas requiring attention, and management considerations**.

3.1 Purpose

The executive summary provides a quick understanding of the project's findings without requiring detailed knowledge of Python, Pandas, or statistical analysis.

3.2 Overall Performance

The executive summary presents the major project-level indicators, including:

- Total Sales
- Gross Profit
- Total Units
- Overall Gross Margin

For the analysed dataset, the overall values identified during the analysis are:

KPI	Result
Total Sales	₹141,783.63
Gross Profit	₹93,442.80
Total Units	38,654
Gross Margin	65.91%

3.3 Division-Level Findings

The executive summary highlights the contribution of the major divisions.

Division	Sales	Gross Profit	Average Margin
Chocolate	₹131,692.90	₹88,824.62	67.46%
Other	₹9,663.25	₹4,333.45	37.67%
Sugar	₹427.48	₹284.73	57.69%

These figures provide stakeholders with a concise view of the relative sales, profit, and margin characteristics of each division.

3.4 Product Concentration

The Pareto analysis indicates that a relatively small number of products account for a substantial proportion of total business performance.

For the analysed dataset:

- **5 out of 15 products** are required to reach approximately 80% of revenue.
- This represents approximately **33.33% of the products**.
- **5 out of 15 products** are required to reach approximately 80% of profit.
- This also represents approximately **33.33% of the products**.

The finding indicates that monitoring key contributing products is important for maintaining overall business performance.

3.5 Margin and Cost Considerations

The analysis identifies products whose margins fall below the selected screening threshold.

These products can be considered for further examination of:

- Pricing
- Discounts
- Procurement cost

- Product-level cost structure
- Sales volume
- Profit contribution

The dashboard provides an interactive mechanism for changing the margin threshold and identifying products requiring attention.

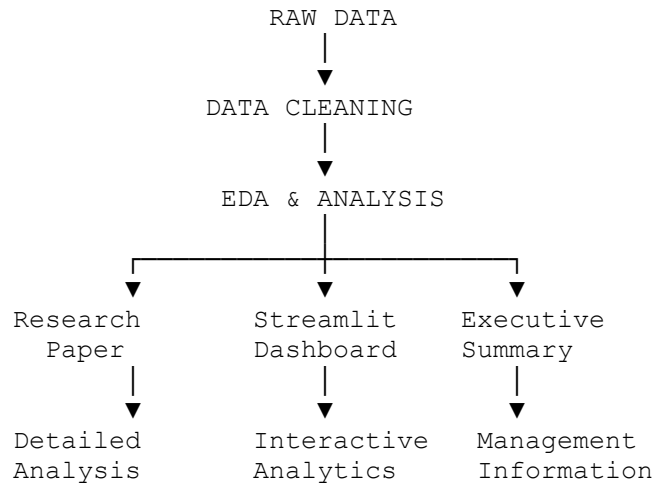
3.6 Management Considerations

The executive summary highlights the following areas for periodic management review:

1. **Key Product Monitoring**
   Products contributing significantly to revenue and profit should be monitored periodically.
2. **Margin Monitoring**
   Products with comparatively lower margins should be reviewed to understand the reasons for lower profitability.
3. **Cost Review**
   Products with relatively high costs should be examined to identify opportunities for procurement or cost-efficiency improvements.
4. **Revenue and Profit Concentration**
   Concentration among a limited number of products should be monitored as part of business-performance review.
5. **Periodic Dashboard Monitoring**
   The Streamlit dashboard can be used as an interactive management-support tool for reviewing changes in profitability and margins.

4. Relationship Between the Three Deliverables

The three deliverables are designed to work together.



Research Paper
Provides the **detailed analytical explanation**.

Streamlit Dashboard
Provides **interactive exploration and live analytical views** of the processed dataset.

Executive Summary
Provides **concise management-level information and key findings**.
Together, these deliverables demonstrate the complete data analytics workflow from **raw data to analysis, visualization, insights, and management-oriented decision support**.